

Exactly solvable problems in the momentum space with a minimum uncertainty in position

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A new approach in solution of simple quantum mechanical problems in deformed space with minimal length is presented. We propose the generalization of Schrödinger equation in momentum representation on the case of deformed Heisenberg algebra with minimal length. Assuming that the kernel of potential energy operator does not change in the case of deformation, we obtain exact solution of eigenproblem of a particle in delta potential as well as double delta potential. Particle in Coulomb like potential is revisited and the problem of invertibility and hermiticity of inverse coordinate operator is solved. *Published by AIP Publishing.* [<http://dx.doi.org/10.1063/1.4945313>]

I. INTRODUCTION

Quantum mechanics with modification of the usual canonical commutation relations has attracted a lot of attention recently. Such interest is motivated by the investigations in string theory and quantum gravity, which suggest the existence of minimal length as a finite lower bound to the possible resolution of length.^{1–3} Minimal length can be achieved by modifying usual canonical commutation relations.^{4–7} In the present paper, we study the simplest deformed algebra

$$[X, P] = i\hbar(1 + \beta P^2) \quad (1)$$

leading to minimum uncertainty in position $\Delta X_{min} = \hbar\sqrt{\beta}$, which is called minimal length.

One can encounter some difficulties connected with the deformation of commutation relations (1) while solving the quantum mechanical problems. Therefore, exact solution of the Schrödinger equation has been obtained only for a few systems. They are the one-dimensional harmonic oscillator with minimal uncertainty in position⁵ and also with minimal uncertainty in both position and momentum,^{8,9} the D-dimensional isotropic harmonic oscillator,^{10,11} the three-dimensional Dirac oscillator,¹² (1+1)-dimensional Dirac oscillator within Lorentz-covariant deformed algebra,¹³ the one dimensional Coulomb problem,^{14,15} and a particle in the singular inverse square potential.^{16,17}

Difficulties in obtaining exact solution of quantum mechanical problems with deformed commutation relation forced to develop the perturbation techniques and numerical calculus. In Ref. 7, the perturbational D-dimensional isotropic harmonic oscillator was considered. Three-dimensional Coulomb problem with deformed Heisenberg algebra was studied within the perturbation theory in the nonrelativistic case^{18–22} and in the case of Lorentz-covariant deformed algebra.^{23,24} Numerical result for hydrogen atom spectrum in a space with deformed commutation relation was obtained in Ref. 19.

The studies of quantum mechanical systems in deformed space with minimal length are interesting not only from the mathematical point of view but also from physical one, because they give the possibility to consider the influence of the deformation on the energy spectra. Comparison of these results with experimental data may yield an estimation of the value of the minimal length (see, for instance, Ref. 20).

In the present paper, we expand the list of exactly solvable quantum mechanical problems in deformed space with minimal length. In Section II, we review Schrödinger equation in momentum representation and generalize it on the case of deformed space. Next, we consider attractive delta potential and double delta potential in Sections III and IV, correspondingly. In Section IV, we revisit one dimensional Coulomb-like problem. Finally, Section V contains the conclusion.

II. SCHRÖDINGER EQUATION IN THE MOMENTUM SPACE

In ordinary quantum mechanics, Schrödinger equation can be written in momentum representation as the following integral equation:

$$\frac{p^2}{2m}\phi(p) + \int_{-\infty}^{\infty} U(p-p')\phi(p')dp' = E\phi(p) \quad (2)$$

with

$$U(p-p') = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} V(x) \exp\left(-\frac{i}{\hbar}(p-p')x\right) dx \quad (3)$$

being the kernel of the potential energy operator. In deformed case, momentum and position operators can be presented using undeformed ones in the form

$$P = \frac{1}{\sqrt{\beta}} \tan \sqrt{\beta} p, \quad (4)$$

$$X = x, \quad (5)$$

with $[x, p] = i\hbar$. We assume the Schrödinger equation in the deformed space reads as follows:

$$\frac{1}{2m\beta} \tan^2(\sqrt{\beta} p) \phi(p) + \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} U(p-p')\phi(p')dp' = E\phi(p). \quad (6)$$

Notice that (3) now is only formally valid, because in deformed space the coordinate representation does not exist due to the minimal length.

III. DELTA POTENTIAL IN DEFORMED SPACE WITH MINIMAL LENGTH

In undeformed case, attractive delta potential in the coordinate representation

$$V(x) = -2\pi\hbar U_0 \delta(x) \quad (7)$$

corresponds to constant potential in momentum space due to (3),

$$U(p-p') = -U_0. \quad (8)$$

We assume that in deformed space, delta potential is still expressed by constant potential in momentum space. The Schrödinger equation then reads

$$\frac{1}{2m\beta} \tan^2(\sqrt{\beta} p) \phi(p) - U_0 \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi(p')dp' = E\phi(p). \quad (9)$$

The solution of (9) can be easily obtained,

$$\phi(p) = \frac{2m\beta U_0 \tilde{\varphi}}{\tan^2(\sqrt{\beta} p) + \beta q^2}. \quad (10)$$

Here we use the notation

$$\tilde{\varphi} = \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi(p')dp' \quad (11)$$

and

$$q = \sqrt{-2mE}. \quad (12)$$

Integrating (10) over p in the range $[-\frac{\pi}{2\sqrt{\beta}}, \frac{\pi}{2\sqrt{\beta}}]$, we obtain

$$q(1 + \sqrt{\beta}q) = 2\pi m U_0, \quad (13)$$

which yields

$$q = \frac{-1 + \sqrt{1 + 8\pi m U_0 \sqrt{\beta}}}{2\sqrt{\beta}}. \quad (14)$$

Here we omit another root of the quadratic equation because it is negative.

The energy spectrum, like in undeformed case, consists of one energy level

$$E = -\frac{1 + 4\pi m U_0 \sqrt{\beta} - \sqrt{1 + 8\pi m U_0 \sqrt{\beta}}}{4m\beta}. \quad (15)$$

For small β energy spectrum can be approximated as

$$E = -2\pi^2 m U_0^2 + 8\pi^3 m^2 U_0^3 \sqrt{\beta} - 40\pi^4 m^3 U_0^4 \beta + o(\beta^{3/2}). \quad (16)$$

The corresponding normalized wavefunction is

$$\phi(p) = \sqrt{\frac{2}{\pi}} \frac{\beta(1 + \sqrt{\beta}q)}{\sqrt{1 + 2\sqrt{\beta}q}} \frac{q^{3/2}}{\tan^2(\sqrt{\beta}p) + \beta q^2}. \quad (17)$$

In the limit $\beta \rightarrow 0$, energy E and corresponding wave function ϕ go to well known undeformed result.

Note that this problem was also considered in the linear approximation over the deformation parameter in Ref. 25. Expression for the energy spectrum obtained in Ref. 25 has other multipliers before $\sqrt{\beta}$ and β in comparison with (16). The reason for this discrepancy of the results is connected with the fact that in Ref. 25, p was mistakenly considered to belong to $(-\infty, \infty)$ instead of $(-\frac{\pi}{2\sqrt{\beta}}, \frac{\pi}{2\sqrt{\beta}})$.

IV. DOUBLE DELTA POTENTIAL

The problem of two equal, symmetrically displaced from the origin of coordinate by distance a , attractive delta potentials is described by

$$V(x) = -\pi\hbar U_0 (\delta(x - a) + \delta(x + a)). \quad (18)$$

This problem corresponds to the harmonic potential in momentum space due to (3),

$$U(p - p') = -U_0 \cos(\alpha(p - p')), \quad (19)$$

with $\alpha = a/\hbar$.

We again assume that in deformed space, this problem is still expressed by potential (19) in momentum space. The Schrödinger equation of the problem can be written as

$$\frac{1}{2m\beta} \tan^2(\sqrt{\beta}p) \phi(p) - U_0 \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \cos(\alpha(p - p')) \phi(p') dp' = E \phi(p). \quad (20)$$

By changing p for $-p$ in (20), we can show that the Hamiltonian commutes with the parity operator. Therefore the Hamiltonian and the parity operator have common system of eigenfunction. Thus eigenfunctions of considerable problem can be chosen as even or odd functions.

Equation (20) can be rewritten as

$$\phi(p) = \frac{m\beta U_0}{\tan^2 \sqrt{\beta}p + \beta q^2} (\tilde{\varphi}_- e^{i\alpha p} + \tilde{\varphi}_+ e^{-i\alpha p}), \quad (21)$$

with $q = \sqrt{-2mE}$ and

$$\tilde{\varphi}_{\pm} = \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} e^{\pm i\alpha p'} \phi(p') dp'. \quad (22)$$

Substituting (21) into (22), we obtain

$$\begin{aligned}\tilde{\varphi}_- &= m\beta U_0 (\tilde{\varphi}_- g(0) + \tilde{\varphi}_+ g(\alpha)), \\ \tilde{\varphi}_+ &= m\beta U_0 (\tilde{\varphi}_- g(\alpha) + \tilde{\varphi}_+ g(0)).\end{aligned}\quad (23)$$

Here we use the notation

$$g(\alpha) = \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \frac{\cos 2\alpha p}{\tan^2 \sqrt{\beta} p + \beta q^2} dp. \quad (24)$$

Since eigenfunctions of considerable problem can be chosen as even or odd functions, we demand the following equality for constants $\tilde{\varphi}_{\pm}$:

$$\tilde{\varphi}_+ = \pm \tilde{\varphi}_-, \quad (25)$$

with upper sign for even and lower one for odd eigenfunctions. Substituting (25) into (21), we obtain eigenfunctions of the problem in the form

$$\phi_{\text{even}}(p) = \frac{A}{\tan^2 \sqrt{\beta} p + \beta q^2} \cos \alpha p, \quad (26)$$

$$\phi_{\text{odd}}(p) = \frac{B}{\tan^2 \sqrt{\beta} p + \beta q^2} \sin \alpha p. \quad (27)$$

By substitution of (25) into (23), we write the equation on the energy spectrum

$$\frac{1}{\beta m U_0} - g(0) = \pm g(\alpha). \quad (28)$$

We failed to obtain analytical expression for integral $g(\alpha)$ in general form. But for some special cases of parameter α , we found

$$g(0) = \frac{\pi}{\beta q(1 + \sqrt{\beta} q)}, \quad (29)$$

$$g(\sqrt{\beta} n) = \frac{\pi(1 - \sqrt{\beta} q)^{n-1}}{\beta q(1 + \sqrt{\beta} q)^{n+1}}, \quad (30)$$

with $n = 1, 2, \dots$

Thus, if the distance between delta wells is $2a = 2n\hbar\sqrt{\beta}$, the equation on the energy spectrum writes

$$\frac{1}{mU_0} - \frac{\pi}{q(1 + \sqrt{\beta} q)} = \pm \frac{\pi(1 - \sqrt{\beta} q)^{n-1}}{q(1 + \sqrt{\beta} q)^{n+1}}. \quad (31)$$

In the limit of $\beta \rightarrow 0$, keeping the distance between delta wells unchanged, the equation on energy spectrum (31) leads to undeformed one²⁶

$$q = \pi m U_0 \left(1 \pm e^{-\frac{2qa}{\hbar}} \right). \quad (32)$$

Note that in the limit $a \rightarrow 0$, we recover the results obtained in Sec. III for the delta potential.

V. COULOMB-LIKE PROBLEM

In undeformed space, the Schrödinger equation of the one dimensional Coulomb-like problem writes

$$\frac{p^2}{2m} \psi(x) - \frac{\alpha}{x} \psi(x) = E \psi(x). \quad (33)$$

We propose the inverse coordinate operator in the following form:

$$\frac{1}{x} = v.p. \frac{1}{x} + A\pi\delta(x), \quad (34)$$

with A being real constant. This definition of the operator $1/x$ corresponds to the following limit:

$$\frac{1}{x} = \lim_{\varepsilon \rightarrow 0} \frac{x + \varepsilon A}{x^2 + \varepsilon^2}. \quad (35)$$

Note that such proposal ensures hermicity of the operator $1/x$. Also in the case of definition (34), the following equality is satisfied:

$$\frac{1}{x} x = x \frac{1}{x} = 1. \quad (36)$$

The kernel of the potential energy operator $V(x) = -\alpha x^{-1}$ in momentum representation reads

$$U(p - p') = -\frac{\alpha}{2\hbar} (2i\theta(p' - p) - i + A), \quad (37)$$

where $\theta(p' - p)$ is the Heaviside step function. We traditionally assume that this kernel remains unchanged in deformed space and write the Schrödinger equation for considerable problem in deformed space with minimal length,

$$\frac{1}{2m\beta} \tan^2(\sqrt{\beta}p) \phi(p) - \frac{\alpha}{2\hbar} \left[(i + A) \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi(p') dp' - 2i \int_{-\frac{\pi}{2\sqrt{\beta}}}^p \phi(p') dp' \right] = E\phi(p). \quad (38)$$

Differentiating the latter integral equation, we obtain the differential one

$$\frac{1}{2m\beta} \left(\tan^2(\sqrt{\beta}p) \phi(p) \right)' + \frac{i\alpha}{\hbar} \phi(p) = E\phi'(p), \quad (39)$$

which yields

$$\phi(p) = \frac{C}{\tan^2(\sqrt{\beta}p) + \beta q^2} \exp \left(-\frac{i\alpha_0}{\sqrt{\beta}(1 - \beta q^2)} \left[\frac{1}{\sqrt{\beta}q} \arctan \frac{\tan(\sqrt{\beta}p)}{\sqrt{\beta}q} - \sqrt{\beta}p \right] \right). \quad (40)$$

Here normalization constant is

$$C = \sqrt{\frac{2}{\pi}} \frac{\beta(1 + \sqrt{\beta}q)q^{3/2}}{\sqrt{1 + 2\sqrt{\beta}q}}. \quad (41)$$

We also use the notation $\alpha_0 = \frac{2m\beta\alpha}{\hbar}$ and $q = \sqrt{-2mE}$.

Notice that from Equation (38), we can extract the definition of inverse position operator

$$\frac{1}{X} \phi(p) = -\frac{i}{\hbar} \int_{-\frac{\pi}{2\sqrt{\beta}}}^p \phi(p') dp' + c[\phi], \quad (42)$$

with $c[\phi]$ denoting the following functional:

$$c[\phi] = \frac{i + A}{2\hbar} \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi(p') dp'. \quad (43)$$

Our definition ensures hermicity of $1/X$ and is in agreement with that proposed in Ref. 14, but here we obtain the explicit form of functional $c[\phi]$, while in Ref. 14 this question was left outside of consideration. It is interesting that for eigenfunctions (40), the following equality is true:

$$\frac{1}{X} X \phi(p) = X \frac{1}{X} \phi(p) = \phi(p). \quad (44)$$

Let us show it straightforwardly. In momentum representation, the position operator is the following: $X = i\hbar \frac{d}{dp}$. Using the definition of the inverse position operator $1/X$ given by (42), we find $X \frac{1}{X} \phi(p) = \phi(p)$. Note that the constant A is eliminated here by differentiating over p . For $\frac{1}{X} X \phi(p)$

we obtain

$$\begin{aligned} \frac{1}{X} X \phi(p) &= \int_{-\frac{\pi}{2\sqrt{\beta}}}^p \frac{d\phi(p')}{dp'} dp' + \frac{iA-1}{2} \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \frac{d\phi(p')}{dp'} dp' = \\ \phi(p) - \phi\left(-\frac{\pi}{2\sqrt{\beta}}\right) &+ \frac{iA-1}{2} \left(\phi\left(\frac{\pi}{2\sqrt{\beta}}\right) - \phi\left(-\frac{\pi}{2\sqrt{\beta}}\right) \right). \end{aligned} \quad (45)$$

Taking into account in the last formula that $\phi(\pm\frac{\pi}{2\sqrt{\beta}}) = 0$, we find $\frac{1}{X} X \phi(p) = \phi(p)$. Note that constant A is removed by multiplying over zero. Thus, Equation (44) holds. Consequently, the problem of invertibility of the position operator X is solved.

The integrals in (38) yield

$$\int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi(p') dp' = \frac{2C}{\alpha_0} \sin \varphi_0, \quad (46)$$

$$\int_{-\frac{\pi}{2\sqrt{\beta}}}^p \phi(p') dp' = \frac{iC}{\alpha_0} \left(\tan^2(\sqrt{\beta}p) \phi(p) - e^{i\varphi_0} \right), \quad (47)$$

with $\varphi_0 = \frac{\pi\alpha_0}{2\beta q(1+\sqrt{\beta}q)}$. Substituting obtained results into Equation (38), we find

$$\sin(\varphi_0 - \delta\pi) = 0 \quad (48)$$

or

$$\frac{m\alpha}{\hbar q(1+\sqrt{\beta}q)} = n + \delta, \quad n = 0, 1, 2, \dots \quad (49)$$

with

$$\delta = \frac{1}{\pi} \operatorname{arccot} A, \quad 0 \leq \delta \leq 1. \quad (50)$$

The endpoints $\delta = 0$ and $\delta = 1$ are reached in the limit of $A \rightarrow +\infty$ and $A \rightarrow -\infty$, correspondingly. The energy spectrum is

$$E_n = -\frac{1}{8m\beta} \left(1 - \sqrt{1 + \frac{4m\alpha}{\hbar(n+\delta)}\sqrt{\beta}} \right). \quad (51)$$

Tending δ to zero for $n = 0$, we obtain that energy goes to $-\infty$ and corresponding eigenfunction vanishes everywhere. This means that for $\delta = 0$, quantum number $n = 1, 2, \dots$ and energy spectra for $\delta = 0$ and $\delta = 1$ coincide. Thus, we obtain the families of spectra, which are the same as presented in Ref. 14. Each of these families is characterized by the value of δ or equivalently A . The values of mentioned parameters can be calculated from the results of an experiment.

In order to explain what happens in the limit $\beta \rightarrow 0$ let us consider the value of the eigenfunction in position representation at the origin. So, for $\psi_n(x)$ corresponding to E_n at the origin, we have

$$\psi_n(x)|_{x=0} = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\frac{\pi}{2\sqrt{\beta}}}^{\frac{\pi}{2\sqrt{\beta}}} \phi_n(p') dp' = \frac{\sqrt{2}C}{\alpha_0\sqrt{\pi\hbar}} \frac{(-1)^n}{\sqrt{1+A^2}}. \quad (52)$$

In the limit of $\beta \rightarrow 0$, the value of the eigenfunction $\psi_n(x)$ at the origin can be written as the following:

$$\psi_n(x)|_{x=0} = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi_n(p') dp' = \frac{\sqrt{\hbar m}}{\pi} \frac{(-1)^n}{\sqrt{1+A^2}} \left(\frac{\alpha}{\hbar(n+\delta)} \right)^{3/2}. \quad (53)$$

The last formula gives a relation of constant A with the value of eigenfunction in the origin of coordinate. Thus, the value of constant A depends on the value of the eigenfunction in position representation at the origin. It is usually assumed in undeformed space that eigenfunction $\psi_n(x)$

vanishes at the origin, which corresponds to the limit of A to infinity. In such a limit functional $c[\phi_n(p)]$ is finite

$$\lim_{A \rightarrow \infty} c[\phi_n(p)] = \frac{(-1)^n \sqrt{m}}{\sqrt{2\pi}} \left(\frac{\alpha}{\hbar n} \right)^{3/2}. \quad (54)$$

Note that the last result is in agreement with the one obtained in Ref. 28.

Besides the fact that our result for energy spectrum coincides with the one obtained in Ref. 14, it is also in agreement with the result obtained in Ref. 15, where different definition of $1/X$ operator was considered. But our result contradicts with the one obtained in Ref. 27 using path integral approach.

VI. CONCLUSION

In this paper, we studied the Schrödinger equation in momentum representation in deformed space with minimal length. Assuming that the kernel of the potential energy does not change in case of deformed commutation relation, we considered Dirac delta potential and revisited Coulomb-like potential in deformed space with minimal length.

In the case of the Dirac delta potential, we obtained that system has one bound state similarly as in undeformed case. We concluded that this system has an interesting property in deformed space. Namely, the first correction to the energy level is proportional to $\sqrt{\beta}$. For previously solved problems, the correction is proportional to β in case of harmonic oscillator, and β or $\beta \ln \beta$ (for s -states) in case of hydrogen atom and $\sqrt{\beta}$ in case of $1D$ potential $1/X$. This fact brings a new possibility to uncover the existence of deformed commutation relations for smaller parameter of deformation.

In case of double delta potential, we obtained the wave functions and the equations for energy levels. In the limit of parameter of deformation to zero, these results coincide with well known undeformed ones. Tending the distance between delta wells to zero, we yield results for the delta potential.

We also proposed the inverse coordinate operator $1/X$ in the form that preserves hermicity and fulfillment of the condition

$$\frac{1}{X} X = X \frac{1}{X} = 1. \quad (55)$$

Our definition of $1/X$ contains one arbitrary real parameter, which means that there exist different extensions of operator $1/X$. We obtain the same families of spectra for the particle in $1D$ potential $1/X$ as in Ref. 14. Each of them is characterized by a value of free parameter. Different values of this parameter correspond to different extensions of operator $1/X$.

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